

# pro\_ask\_count\_injection — review copy

the GPT-5.5 ask + the verbatim Isabelle definitions it ships with

Secretary working note — 2026-06-19

This PDF is for human review of the self-contained GPT ask in `posix-codex/pro_ask_count_injection/`. It reproduces, *verbatim and unedited*, the two files the external model receives: `PROMPT.txt` (the ask) and `DEFINITIONS.txt` (every function + cited green lemma, copied from the Isabelle source with `file:line`). The third attachment `CARD_FRONTIER.md` (the one-page state summary) is plain markdown and is not reproduced here.

## 1. The ask — `PROMPT.txt`

```
ROLE: Proof architect for an Isabelle/HOL cubic size-bound proof of a POSIX regex lexer. The WHOLE Gate is
  ↪ GREEN
(0 sorry) except ONE linear row-COUNT lemma, now pinned to a single cross-prune step. I need a CONCRETE
  ↪ DESIGN that
closes it (or a precise argument that no such design exists, and what a correct proof must track).

YOU DO NOT NEED TO WRITE OR MACHINE-CHECK ISABELLE. We do the Isabelle formalization ourselves. Your
  ↪ deliverable is:
(1) the DESIGN (the injection / charge / invariant, in plain math + prose); (2) the precise RECURSIVE
  ↪ DEFINITIONS of
every new function you introduce (in the same total-recursive style as DEFINITIONS.txt, by cases on the
  ↪ rregexp
constructors, so we can transcribe them directly); (3) PYTHON EVIDENCE that it has no counterexamples (see
  ↪ §0). The
verification you CAN and MUST do is the Python check -- do not hand back an untested idea.

EVERY function and cited lemma below is given VERBATIM (with file:line) in the attached DEFINITIONS.txt.
  ↪ Use those
exact definitions -- do NOT work from approximate/remembered semantics (a prior attempt failed precisely
  ↪ by mis-modeling
the collapsing tail). The state summary is in the attached CARD_FRONTIER.md.

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0. PROTOTYPE & SELF-VALIDATE IN PYTHON BEFORE YOU ANSWER (mandatory)
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You have a Python tool -- USE IT. Do NOT return a paper proof you have not run: the previous external
  ↪ attempt failed
precisely because its construction (a "collapsing-tail" ledger) was never tested and was false on a size
  ↪ -12 regex.
Procedure:
1. Build a FAITHFUL Python model of the functions in DEFINITIONS.txt (use the EXACT equations). At
  ↪ minimum: the
  rregexp datatype (ZERO/ONE/CHAR/SEQ/ALTS/STAR suffice on the clean fragment); rsize; rsimp4_SEQ_atom (
  ↪ sigma4);
  rsimp7_SEQ_atom (sigma7 -- collapses ONLY a leading a*.a*); rflts; rdistinct; rsimp_ALTs; the prune
  ↪ chain
  (rprune_eq_against -> rsimpStrong_prune_pair_raw -> rsimpStrong_prune_against_rows_raw ->
  rsimpStrong_prune_rows_acc_raw -> rsimpStrong_prune_rows_raw); rsimpStrong_raw (S); rfrontier;
  ↪ row_dlforms (dl);
  rsimpStrong_dlform_closure; apder_terms; apder_frontier; apder_rows; apder_term_frontier_acc;
  ↪ strong_apder_acc;
  apder_strong_dlfrontier. SANITY-CHECK the model: card(apder_strong_dlfrontier r) <= rsize r + 1 must
  ↪ hold with
  0 violations, and the named CEs below must reproduce in your model. If your model disagrees, fix the
  ↪ model first.
2. Encode your proposed injection / charge / invariant as Python and TEST it on: (a) exhaustive small
  ↪ regexes
```

(rsize up to ~10); (b) the witness family -- nested  $\text{SEQ}(\text{ALTS}[\text{pre}, 1], \dots \text{SEQ}(\text{atom}, \text{star}^*))$  opened at  $\hookrightarrow k = \text{star}^*$ , and stars-against-stars; (c) these named REGRESSION counterexamples (encode them and confirm your  $\hookrightarrow$  construction survives each):

- SAA RALTS leak (rsize 7):  $r = \text{ALTS}[\text{ONE}, \text{SEQ}(\text{STAR } b, \text{STAR } a)]$ ,  $k = \text{STAR } a$ .  
(strong\_apder\_acc (ALTS rs) k keeps up to 4 uncollapsed ...( $s \cdot s$ ) rows present in NO child.)
- collapsing-tail over-approx CE (rsize 12):  
 $r = \text{SEQ}(\text{ALTS}[\text{ONE}, \text{SEQ}(\text{ALTS}[\text{CHAR } a, \text{ONE}], \text{STAR } a), \text{CHAR } a], \text{STAR } a)$   
( $U(r)$  contains the UNCOLLAPSED rows  $a \cdot a$  and  $a \cdot (a \cdot a)$  -- a re-running-S tail loses them).
- Tail2 unreachable-continuation leak: the SEQ step holds on REACHABLE k but leaks at standalone k  
 $\hookrightarrow = \text{SEQ}(\text{STAR } b, \text{STAR } b)$ .
- awidth refutation:  $r = \text{ALTS}[\text{ONE}, \text{STAR}(\text{ALTS}[\text{STAR}(\text{CHAR } c), \text{ONE}])]$  has  $\text{card}(U)=3 > \text{awidth}+1=2$  (so do  $\hookrightarrow$  NOT target awidth+1).

3. ITERATE: when your construction has a counterexample, find the SMALLEST one, diagnose, and REVISE --  $\hookrightarrow$  repeat until  
0 violations across (a)+(b)+(c) AND (for an injection) injectivity holds with 0 collisions.

4. ONLY THEN write up the design (§3). In your answer, REPORT the Python validation: the exact predicate  $\hookrightarrow$  (s) you tested, the sample space/size, that it found 0 counterexamples (incl. the named CEs), and any case  $\hookrightarrow$  your model could not cover. A design you did not run in Python will be rejected. (Include the Python code, or  $\hookrightarrow$  enough of it that we can re-run it.)

## 1. THE OPEN LEMMA

```
lemma card_apder_strong_dlfntier_le:
  assumes "apder_clean r"
  shows "card (apder_strong_dlfntier r) \<le> Suc (rsize r)"      (* a LINEAR row COUNT *)
Write U(r) := apder_strong_dlfntier r = rsimpStrong_dlform_closure (apder_rows r)
      = (UN q : apder_rows r. row_dlforms (rsimpStrong_raw q)).
Any LINEAR bound (card (U r) <= a * rsize r + b) also closes the Gate (loosen the downstream budget). A
   $\hookrightarrow$  QUADRATIC
bound does NOT (it makes the Gate quartic). Discharging this lemma feeds
   $\hookrightarrow$  actual_gate_from_direct_universe_rowlevel
(DEFINITIONS.txt §D) and the cubic Gate becomes unconditional. The bound is machine-validated TRUE (0
   $\hookrightarrow$  violations,
tightest slack 0).
```

## 2. THE WALL (validated; do NOT re-propose a clean subset)

Three independent prover agents converged on this. The clean-subset induction is dead on BOTH natural  $\hookrightarrow$  carriers, each leaking at exactly ONE constructor via the SAME  $a \cdot a$  cross-prune (the SET-level rsimpStrong\_ALTs\_raw  $\hookrightarrow$  prune inside rsimpStrong\_raw keeps uncollapsed rows like ...( $s \cdot s$ ) that a per-row S would collapse):

- \*  $U(r) = \text{apder\_strong\_dlfrontier } r$  is RALTS-clean ( $\text{apder\_strong\_dlfrontier\_RALTS\_subset}$ , GREEN) but RSEQ  $\hookrightarrow$  /RSTAR-LEAKY.
- \*  $\text{SAA}(r, k) = \text{strong\_apder\_acc } r \ k$  (the strong closure of the sigma4 carrier; it DISTRIBUTES over its  $\hookrightarrow$  carrier set, so RCHAR/RSEQ/RSTAR carrier splits are CLEAN SUBSETS -- see  $\text{strong\_apder\_acc}\{\text{RSEQ}, \text{RSTAR}, \text{RCHAR}\}\text{\_subset} \hookrightarrow$  , GREEN) but is RALTS-under-continuation LEAKY:  $\text{strong\_apder\_acc } (\text{RALTS } rs) \ k$  is NOT SUBSET of  $(\text{UN } q. \hookrightarrow \text{strong\_apder\_acc } q \ k)$  (smallest CE  $r=(1 + b \cdot a)$ ,  $k=a$ ; the prune keeps up to 4 uncollapsed ...( $s \cdot s$ ) rows present in no  $\hookrightarrow$  child).

So no single carrier closes by card\_mono on a subset.

REFUTED routes (do NOT re-propose): the false  $\text{dl}(S \ q) \text{ SUBSET } S' \text{dl } q$ ;  $U \ r \text{ SUBSET } S' \text{Uun } r$  (~56/66833 leak); per-row size bounds (quartic); the deep frontier ( $U$  is NOT a subset of it); the awidth+1 injection (FALSE:  $\hookrightarrow r=(1+(c+1) \cdot)$  gives  $\text{card}(U)=3 > \text{awidth}+1=2$ , every position key collides); the opened\_boundary\_forms transfer (it is  $\hookrightarrow$  sigma4 / NON-strong -- see the GREEN refutation rsimpStrong\_dlform\_closure\_opened\_boundary\_carrier\_false in DEFINITIONS.txt §E  $\hookrightarrow$  ); the rewrite

route ->r' (does not commute with the derivative without the full ~1000-line Ch5 rewrite system).

### 3. WHAT I WANT (pick the SAA carrier; 3 of 4 constructor diff-steps already clean)

On the SAA carrier the diff-card induction  $\text{card}(\text{strong\_apder\_acc } r \text{ } k - \text{strong\_apder\_acc } \text{RONE } k) \leq \text{rsize } r$   
→ has CLEAN,  
already-written steps for RCHAR ( $\text{card\_strong\_apder\_acc\_RCHAR\_diff\_base\_le}$ ), RSEQ (via  
→  $\text{card\_Un\_Diff\_telescope\_le} +$   
 $\text{strong\_apder\_acc\_RSEQ\_subset} + \text{strong\_apder\_acc\_RONE\_sigma\_subset}$ ), and RSTAR ( $\text{telescope} + \text{a root-}\leq -1$   
→ lemma). The ONE  
remaining step is the LEAKY RALTS one, which has NO clean subset:

GOAL STEP:  $\text{card}(\text{strong\_apder\_acc } (\text{RALTS } rs) \text{ } k - \text{strong\_apder\_acc } \text{RONE } k)$   
 $\leq (\text{SUM}_{\{q \text{ in set } rs\}} \text{card}(\text{strong\_apder\_acc } q \text{ } k - \text{strong\_apder\_acc } \text{RONE } k)) + 1.$

CORRECTION (2026-06-19, after a Pro round + our re-validation): the "+1" form above is FALSE -- for n

→ independent  
branches  $q_i = (x_i \cdot a^*)$  under  $k = a^*$ , the parent has  $2n$  non-base rows but the child sum is  $n$ , so  $2n \leq n$   
→ +1 fails for  
 $n \geq 2$  (CE:  $rs = [b \cdot a^*, c \cdot a^*]$ ,  $k = a^*$ , LHS 4 > RHS 3). The per-branch tail-collapse  $(x \cdot a^*) \cdot a^* \rightarrow x \cdot (a \cdot a^*)$   
→ needs a

LINEAR per-branch debt, not a single +1. BUT the route is NOT dead: the GLOBAL bound  
 $\text{card}(\text{strong\_apder\_acc } r \text{ } \text{RONE} - \text{strong\_apder\_acc } \text{RONE } \text{RONE}) \leq \text{rsize } r$  HOLDS WITH ~5x SLACK (the doubling  
→ only occurs

at intermediate continuations, not at the top  $k = \text{RONE}$ ). So the obstruction is the per-step DECOMPOSITION,  
→ not the

truth of the count. Give EITHER:

- (a) a definable injection / transport LEDGER from the LHS prune-residual rows to the per-child RHS  
→ counts (+1 slack),  
operating at the BRANCH-REGEX level (transport a strong row to its SOURCE BRANCH under the same  
→ continuation,  
BEFORE opening -- analogous to a branch-origin lemma for  $\text{rsimpStrong\_ALTs\_raw}$ : every opened row of a  
→ SET-pruned  
strong alternation has a branch origin), accounting for the cross-pruned  $\dots(s \cdot s^*)$  survivors; OR
- (b) a "reachable clean-plug continuation" invariant  $P(k)$  that holds of every  $k$  the induction actually  
→ reaches  
(top  $k = \text{RONE}$ , then  $k := \text{rsimp4\_SEQ\_atom } r2 \text{ } k / \text{rsimp4\_SEQ\_atom } (\text{RSTAR } r) \text{ } k$  down the recursion) and  
→ is strong  
enough that the RALTS step holds under  $P(k)$  -- the leak is ONLY on UNREACHABLE standalone  
→ continuations  
(e.g.  $k = b \cdot b^*$ ), so a precise reachability invariant threaded through the induction would avoid it;  
→ OR
- (c) a proof that NO such injection/invariant is definable, with the minimal extra structure a correct  
→ proof must  
carry (a prune-order index? a provenance tag on rows?).

Deliver, concretely:

- (1) DESIGN: the injection / charge / invariant, in plain math + prose -- what maps to what, why  
→ injective/why the  
+1 slack suffices, how the cross-pruned  $\dots(s \cdot s^*)$  survivors are accounted for. Name the induction  
→ variable and  
the continuation parameter, and say which DEFINITIONS.txt §D green facts each step would use.
- (2) RECURSIVE DEFINITIONS: for every NEW function you introduce (e.g. the ledger/charge map, an  
→ injection key, or a  
reachable-continuation predicate  $P(k)$ ), give its full recursive definition by cases on the rregexp  
→ constructors  
(RZERO/RONE/RCHAR/RSEQ/RALTS/RSTAR), total, in the SAME style as DEFINITIONS.txt -- so we can  
→ transcribe it  
directly into Isabelle. (Pseudocode / math is fine; precise Isabelle syntax is welcome but NOT  
→ required.)
- (3) PYTHON EVIDENCE (§0): your model + the validated predicate(s), 0 counterexamples incl. the named CEs  
→ .

We then formalize it. Once the RALTS step holds, the induction closes and  $\text{card}(U \text{ } r) \leq \text{Suc}(\text{rsize } r)$

→ follows via  
 $\text{apder\_strong\_dlfrontier\_subset\_strong\_apder\_acc\_RONE} + \text{card\_le\_Suc\_card\_Diff\_singleton}$ , and the Gate via  
 $\text{actual\_gate\_from\_direct\_universe\_rowlevel}$  -- so you only need the RALTS step (or whichever of (a)/(b)/(c)  
→ you take).

ATTACHMENTS: DEFINITIONS.txt (verbatim defs + green-lemma statements -- READ THIS), CARD\_FRONTIER.md (  
     $\hookrightarrow$  state summary).

## 2. Verbatim definitions + green-lemma statements — DEFINITIONS.txt

VERBATIM ISABELLE DEFINITIONS + GREEN-LEMMA STATEMENTS

=====

Every function/lemma referenced by PROMPT.txt, copied verbatim from the source (file:line given).  
 Regex datatype rrex constructors: RZERO, RONE, RCHAR c, RSEQ r1 r2, RALTS rs (rs :: rrex list),  
 RSTAR r, RNTIMES r n, RBACKREF4 .., RHALF .., RRESIDUE .. . On the clean fragment (apder\_clean) only  
 RZERO/RONE/RCHAR/RSEQ/RALTS/RSTAR occur (RNTIMES/backref-family excluded by rntimes\_free + apder\_nf).

OPAQUE (you do NOT need their internals): 'afactored1 r s' and 'rpder\_strong\_rows\_raw c rs' are the lexer'

↪ s

derivative-rows functions; they appear ONLY in the final Gate corollary

↪ actual\_gate\_from\_direct\_universe\_rowlevel

(SD), which is a GREEN black-box that consumes your count bound. Your target lemma

↪ card\_apder\_strong\_dlfrontier\_le

does NOT mention them. So treat them as opaque; prove only the count, and the Gate follows.

=====

A. CORE SIMPLIFIER / SEQUENCING FUNCTIONS

(\* base/BasicIdentities.thy:487 \*)

fun rsize :: "rrex \<Rightarrow> nat" where

```
"rsize RZERO = 1"
| "rsize (RONE) = 1"
| "rsize (RCHAR c) = 1"
| "rsize (RALTS rs) = Suc (sum_list (map rsize rs))"
| "rsize (RSEQ r1 r2) = Suc (rsize r1 + rsize r2)"
| "rsize (RSTAR r) = Suc (rsize r)"
| "rsize (RNTIMES r n) = Suc (rsize r) + n"
| "rsize (RBACKREF4 r1 r2 r3 r4 cs) = Suc (rsize r1 + rsize r2 + rsize r3 + rsize r4)"
| "rsize (RHALF r cs rep) = Suc (rsize r)"
| "rsize (RRESIDUE cs rep) = 1"
```

(\* base/BasicIdentities.thy:287 -- the WEAK sequence plug "sigma4" (NO a\*.a\* collapse) \*)

fun rsimp4\_SEQ\_atom :: "rrex \<Rightarrow> rrex \<Rightarrow> rrex" where

```
"rsimp4_SEQ_atom RZERO r2 = RZERO"
| "rsimp4_SEQ_atom RONE r2 = r2"
| "rsimp4_SEQ_atom (RSEQ r1 r2) r3 = rsimp4_SEQ_atom r1 (rsimp4_SEQ_atom r2 r3)"
| "rsimp4_SEQ_atom (RCHAR c) RZERO = RZERO"
| "rsimp4_SEQ_atom (RCHAR c) RONE = RCHAR c"
| "rsimp4_SEQ_atom (RCHAR c) r2 = RSEQ (RCHAR c) r2"
| "rsimp4_SEQ_atom (RALTS rs) RZERO = RZERO"
| "rsimp4_SEQ_atom (RALTS rs) RONE = RALTS rs"
| "rsimp4_SEQ_atom (RALTS rs) r2 = RSEQ (RALTS rs) r2"
| "rsimp4_SEQ_atom (RSTAR r) RZERO = RZERO"
| "rsimp4_SEQ_atom (RSTAR r) RONE = RSTAR r"
| "rsimp4_SEQ_atom (RSTAR r) r2 = RSEQ (RSTAR r) r2"
| "rsimp4_SEQ_atom (RNTIMES r n) RZERO = RZERO"
| "rsimp4_SEQ_atom (RNTIMES r n) RONE = RNTIMES r n"
| "rsimp4_SEQ_atom (RNTIMES r n) r2 = RSEQ (RNTIMES r n) r2"
| "rsimp4_SEQ_atom (RBACKREF4 r1 r2 r3 r4 cs) RZERO = RZERO"
| "rsimp4_SEQ_atom (RBACKREF4 r1 r2 r3 r4 cs) RONE = RBACKREF4 r1 r2 r3 r4 cs"
| "rsimp4_SEQ_atom (RBACKREF4 r1 r2 r3 r4 cs) r5 = RSEQ (RBACKREF4 r1 r2 r3 r4 cs) r5"
| "rsimp4_SEQ_atom (RHALF r cs rep) RZERO = RZERO"
| "rsimp4_SEQ_atom (RHALF r cs rep) RONE = RHALF r cs rep"
| "rsimp4_SEQ_atom (RHALF r cs rep) r2 = RSEQ (RHALF r cs rep) r2"
| "rsimp4_SEQ_atom (RRESIDUE cs rep) RZERO = RZERO"
| "rsimp4_SEQ_atom (RRESIDUE cs rep) RONE = RRESIDUE cs rep"
| "rsimp4_SEQ_atom (RRESIDUE cs rep) r2 = RSEQ (RRESIDUE cs rep) r2"
```

(\* base/BasicIdentities.thy:414 -- the STRONG plug "sigma7": sigma4 + top-head a\*.a\* -> a\* collapse ONLY

↪ \*)

definition rsimp7\_SEQ\_atom :: "rrex \<Rightarrow> rrex \<Rightarrow> rrex" where

```
"rsimp7_SEQ_atom r1 r2 =
  (case (r1, r2) of
    (RSTAR r, RSTAR s) \<Rightarrow> if r = s then RSTAR r else rsimp4_SEQ_atom r1 r2
  | (RSTAR r, RSEQ (RSTAR s) k) \<Rightarrow> if r = s then RSEQ (RSTAR r) k else rsimp4_SEQ_atom r1 r2
```

```

| _ \<Rightarrow> rsimp4_SEQ_atom r1 r2)"

(* base/GeneralRegexBound.thy:18365 -- the strong normaliser "S". RSEQ via sigma7; RALTS via the SET-
  ↪ level
  rsimpStrong_ALTs_raw (the cross-row prune); RSTAR idempotent. *)
fun rsimpStrong_raw :: "rrexpr \<Rightarrow> rrexpr" where
  "rsimpStrong_raw RZERO = RZERO"
| "rsimpStrong_raw RONE = RONE"
| "rsimpStrong_raw (RCHAR c) = RCHAR c"
| "rsimpStrong_raw (RSEQ r1 r2) = rsimp7_SEQ_atom (rsimpStrong_raw r1) (rsimpStrong_raw r2)"
| "rsimpStrong_raw (RALTS rs) = rsimpStrong_ALTs_raw (rflts (map rsimpStrong_raw rs))"
| "rsimpStrong_raw (RSTAR r) =
  (case rsimpStrong_raw r of
    RZERO \<Rightarrow> RONE | RONE \<Rightarrow> RONE | RSTAR s \<Rightarrow> RSTAR s | s \<Rightarrow>
    ↪ RSTAR s)"
| "rsimpStrong_raw (RNTIMES r n) = RNTIMES (rsimpStrong_raw r) n"
| "rsimpStrong_raw (RBACKREF4 r1 r2 r3 r4 cs) = RBACKREF4 r1 r2 r3 r4 cs"
| "rsimpStrong_raw (RHALF r cs rep) = RHALF r cs rep"
| "rsimpStrong_raw (RRESIDUE cs rep) = RRESIDUE cs rep"

(* base/GeneralRegexBound.thy:18264 -- the SET-level alternation prune (rsimpStrong_prune_rows_raw is the
  cross-row pruning scan; THIS is the source of the "cross-prune keeps uncollapsed ... (s*.s*)" survivors)
  ↪ . *)
definition rsimpStrong_ALTs_raw :: "rrexpr list \<Rightarrow> rrexpr" where
  "rsimpStrong_ALTs_raw rs =
    rsimp_ALTs (rdistinct (rflts (rsimpStrong_prune_rows_raw rs)) {})"

(* base/BasicIdentities.thy:206 *)
fun rsimp_ALTs :: "rrexpr list \<Rightarrow> rrexpr" where
  "rsimp_ALTs [] = RZERO"
| "rsimp_ALTs [r] = r"
| "rsimp_ALTs rs = RALTS rs"

(* base/BasicIdentities.thy:177 -- flatten nested alternations, drop RZERO *)
fun rflts :: "rrexpr list \<Rightarrow> rrexpr list" where
  "rflts [] = []"
| "rflts (RZERO # rs) = rflts rs"
| "rflts ((RALTS rs1) # rs) = rs1 @ rflts rs"
| "rflts (r1 # rs) = r1 # rflts rs"

(* base/BasicIdentities.thy:83 -- dedup keeping first occurrence (acc = seen set) *)
fun rdistinct :: "'a list \<Rightarrow> 'a set \<Rightarrow> 'a list" where
  "rdistinct [] acc = []"
| "rdistinct (x#xs) acc = (if x \<in> acc then rdistinct xs acc else x # (rdistinct xs ({x} \<union> acc))
  ↪ )"

(* base/BasicIdentities.thy:25 *)
fun rnullable :: "rrexpr \<Rightarrow> bool" where
  "rnullable (RZERO) = False"
| "rnullable (RONE) = True"
| "rnullable (RCHAR c) = False"
| "rnullable (RALTS rs) = (\<exists>r \<in> set rs. rnullable r)"
| "rnullable (RSEQ r1 r2) = (rnullable r1 \<and> rnullable r2)"
| "rnullable (RSTAR r) = True"
| "rnullable (RNTIMES r n) = (if n = 0 then True else rnullable r)"
| "rnullable (RBACKREF4 r1 r2 r3 r4 cs) = (rnullable r1 \<and> rnullable r2 \<and> rnullable r3 \<and>
  ↪ rnullable r4 \<and> cs = [])"
| "rnullable (RHALF r cs rep) = (rnullable r \<and> cs = [])"
| "rnullable (RRESIDUE cs rep) = (cs = [])"

-- the cross-row pruning scan used inside rsimpStrong_ALTs_raw, VERBATIM (5 functions, dependency order):

(* base/GeneralRegexBound.thy:17015 -- branch-set difference: keep the rrs-branches NOT in lrs (covered =
  ↪ lrs) *)
fun rprune_eq_against :: "rrexpr list \<Rightarrow> rrexpr list \<Rightarrow> rrexpr list" where
  "rprune_eq_against covered [] = []"
| "rprune_eq_against covered (r # rs) =
  (if r \<in> set covered

```

```

    then rprune_eq_against covered rs
    else r # rprune_eq_against covered rs)"

(* base/GeneralRegexBound.thy:18163 -- the PAIRWISE rule: fires ONLY when both rows are (RALTS lrs).k1
   ↪ and
   (RALTS rrs).k2 with the SAME right factor k1 = k2; deletes from the later row's head-alternation the
   ↪ branches
   already in the earlier one, then re-plugs via sigma7. Every other pair leaves the LATER row unchanged.
   ↪ *)
definition rsimpStrong_prune_pair_raw :: "rrexpr \<Rightarrow> rrexpr \<Rightarrow> rrexpr" where
  "rsimpStrong_prune_pair_raw earlier later =
    (case (earlier, later) of
      (RSEQ (RALTS lrs) k1, RSEQ (RALTS rrs) k2) \<Rightarrow>
        if k1 = k2
        then rsimp7_SEQ_atom (rsimp_ALTs (rprune_eq_against lrs rrs)) k2
        else later
    | _ \<Rightarrow> later)"

(* base/GeneralRegexBound.thy:18227 -- prune ONE row against ALL earlier rows (fold the pairwise rule
   ↪ over seen) *)
fun rsimpStrong_prune_against_rows_raw :: "rrexpr list \<Rightarrow> rrexpr \<Rightarrow> rrexpr" where
  "rsimpStrong_prune_against_rows_raw [] r = r"
| "rsimpStrong_prune_against_rows_raw (x # xs) r =
  rsimpStrong_prune_against_rows_raw xs (rsimpStrong_prune_pair_raw x r)"

(* base/GeneralRegexBound.thy:18233 -- left-to-right scan; the accumulator 'seen' collects the PRUNED
   ↪ rows r'
   (NOT the originals), and rows are shrunk-never-dropped (length-preserving). *)
fun rsimpStrong_prune_rows_acc_raw :: "rrexpr list \<Rightarrow> rrexpr list \<Rightarrow> rrexpr list" where
  "rsimpStrong_prune_rows_acc_raw seen [] = []"
| "rsimpStrong_prune_rows_acc_raw seen (r # rs) =
  (let r' = rsimpStrong_prune_against_rows_raw seen r
   in r' # rsimpStrong_prune_rows_acc_raw (r' # seen) rs)"

(* base/GeneralRegexBound.thy:18239 -- entry point (start with empty seen) *)
definition rsimpStrong_prune_rows_raw :: "rrexpr list \<Rightarrow> rrexpr list" where
  "rsimpStrong_prune_rows_raw rs = rsimpStrong_prune_rows_acc_raw [] rs"

(* KEY PROPERTY (why this is the wall): the pairwise rule only DELETES branches already present in an
   ↪ earlier row
   (rows shrunk, never dropped; language-preserving; size-non-increasing). The deletion is at the head-
   ↪ alternation
   SET level and the re-plug uses sigma7, which collapses ONLY a leading a*.a*. So a row of the form ...(s
   ↪ *.s*)
   whose head is NOT a leading a* can SURVIVE the prune UNCOLLAPSED -- this cross-prune residual has no
   ↪ per-row-S
   pre-image, which is exactly why U(SEQ ..)/SAA(RALTS ..) admit no clean subset. *)

=====
B. OPENING / FRONTIER / STRONG UNIVERSE
=====

(* base/GeneralRegexBound.thy:3862 -- the top-alternation split *)
fun rfrontier :: "rrexpr \<Rightarrow> rrexpr set"
  and rfrontiers :: "rrexpr list \<Rightarrow> rrexpr set" where
  "rfrontier RZERO = {}"
| "rfrontier (RALTS rs) = rfrontiers rs"
| "rfrontier r = {r}"
| "rfrontiers [] = {}"
| "rfrontiers (r # rs) = rfrontier r \<union> rfrontiers rs"

(* active/AntimirovFactoredTransition.thy:4791 -- "dl": open one regex to its linear rows.
   Distributes a top alternation; an alternation-headed sequence distributes its branches over the tail
   via sigma7 (the ONLY other place a*.a* can collapse); else bottoms out at the frontier. *)
function (sequential) row_dlforms :: "rrexpr \<Rightarrow> rrexpr set" where
  "row_dlforms RZERO = {}"
| "row_dlforms (RALTS rs) = (\<Union>q \<in> set rs. row_dlforms q)"
| "row_dlforms (RSEQ (RALTS ps) k) = (\<Union>p \<in> set ps. row_dlforms (rsimp7_SEQ_atom p k))"

```

```

| "row_dlforms r = rfrontier r"
(* termination by measure rsize *)

(* active/AntimirovFactoredTransition.thy:5373 *)
definition row_dlformss :: "rrexpr list \<Rightarrow> rrexpr set" where
  "row_dlformss rs = (\<Union>q \<in> set rs. row_dlforms q)"

(* active/AntimirovFactoredTransition.thy:5376 *)
definition row_dlformss_set :: "rrexpr set \<Rightarrow> rrexpr set" where
  "row_dlformss_set U = (\<Union>q \<in> U. row_dlforms q)"

(* active/AntimirovFactoredTransition.thy:11335 -- the STRONG-opened closure of a set: open dl AFTER S.
NB: S is applied INSIDE the union (per member), which is exactly why it DISTRIBUTES over its carrier
  \<=> set. *)
definition rsimpStrong_dlform_closure :: "rrexpr set \<Rightarrow> rrexpr set" where
  "rsimpStrong_dlform_closure U = (\<Union>p \<in> U. row_dlforms (rsimpStrong_raw p))"

(* active/AntimirovFactoredTransition.thy:11400 -- U(r), the target universe (over the deduped Antimirov
  \<=> rows) *)
definition apder_strong_dlfrontier :: "rrexpr \<Rightarrow> rrexpr set" where
  "apder_strong_dlfrontier r = rsimpStrong_dlform_closure (apder_rows r)"

=====
C. ANTIMIROV STRUCTURES + REGIME PREDICATES
=====

(* active/AntimirovFactoredTransition.thy:1715 -- Antimirov partial-derivative TERMS *)
fun apder_terms :: "rrexpr \<Rightarrow> rrexpr set" where
  "apder_terms RZERO = {}"
| "apder_terms RONE = {}"
| "apder_terms (RCHAR c) = {RONE}"
| "apder_terms (RALTS rs) = (\<Union>q \<in> set rs. apder_terms q)"
| "apder_terms (RSEQ r1 r2) = ((\<lambda>p. rsimp4_SEQ_atom p r2) ' apder_terms r1 \<union> apder_terms r2
  \<=> )"
| "apder_terms (RSTAR r) = ((\<lambda>p. rsimp4_SEQ_atom p (RSTAR r)) ' apder_terms r)"
| "apder_terms (RNTIMES r n) = (\<Union>m \<in> {...n}. ((\<lambda>p. rsimp4_SEQ_atom p (RNTIMES r m)) '
  \<=> apder_terms r))"
| "apder_terms (RBACKREF4 r1 r2 r3 r4 cs) = {}"
| "apder_terms (RHALF r cs rep) = {}"
| "apder_terms (RRESIDUE cs rep) = {}"

(* active/AntimirovFactoredTransition.thy:1732 *)
definition apder_frontier :: "rrexpr \<Rightarrow> rrexpr set" where
  "apder_frontier r = rfrontier r \<union> (\<Union>q \<in> apder_terms r. rfrontier q)"

(* active/AntimirovFactoredTransition.thy:1736 -- the deduped Antimirov ROWS (index set of U(r)) *)
definition apder_rows :: "rrexpr \<Rightarrow> rrexpr set" where
  "apder_rows r = insert r (apder_frontier r)"

(* active/AntimirovFactoredTransition.thy:1739 -- the term-frontier ACCUMULATOR (continuation-parametric;
the carrier strong_apder_acc is its strong closure). This is the D-law's accumulator. *)
fun apder_term_frontier_acc :: "rrexpr \<Rightarrow> rrexpr \<Rightarrow> rrexpr set" where
  "apder_term_frontier_acc RZERO k = {}"
| "apder_term_frontier_acc RONE k = {}"
| "apder_term_frontier_acc (RCHAR c) k = rfrontier k"
| "apder_term_frontier_acc (RALTS rs) k = (\<Union>q \<in> set rs. apder_term_frontier_acc q k)"
| "apder_term_frontier_acc (RSEQ r1 r2) k =
  apder_term_frontier_acc r1 (rsimp4_SEQ_atom r2 k) \<union> apder_term_frontier_acc r2 k"
| "apder_term_frontier_acc (RSTAR r) k = apder_term_frontier_acc r (rsimp4_SEQ_atom (RSTAR r) k)"
| "apder_term_frontier_acc (RNTIMES r n) k =
  (\<Union>m \<in> {...n}. apder_term_frontier_acc r (rsimp4_SEQ_atom (RNTIMES r m) k))"
| "apder_term_frontier_acc (RBACKREF4 r1 r2 r3 r4 cs) k = {}"
| "apder_term_frontier_acc (RHALF r cs rep) k = {}"
| "apder_term_frontier_acc (RRESIDUE cs rep) k = {}"

(* active/AntimirovFactoredTransition.thy:1779 -- alphabetic width (counts only RCHAR positions) *)
fun apder_ewidth :: "rrexpr \<Rightarrow> nat" where
  "apder_ewidth RZERO = 0"

```



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| "apder_ewidth RONE = 0"
| "apder_ewidth (RCHAR c) = 1"
| "apder_ewidth (RALTS rs) = sum_list (map apder_ewidth rs)"
| "apder_ewidth (RSEQ r1 r2) = apder_ewidth r1 + apder_ewidth r2"
| "apder_ewidth (RSTAR r) = apder_ewidth r"
| "apder_ewidth (RNTIMES r n) = n * apder_ewidth r"
| "apder_ewidth (RBACKREF4 r1 r2 r3 r4 cs) = 0"
| "apder_ewidth (RHALL r cs rep) = 0"
| "apder_ewidth (RRESIDUE cs rep) = 0"

(* active/AntimirovFactoredTransition.thy:1791 -- the structural normal form *)
fun apder_nf :: "rrexpr \<Rightarrow> bool" where
  "apder_nf RZERO = True"
| "apder_nf RONE = True"
| "apder_nf (RCHAR c) = True"
| "apder_nf (RALTS rs) = (\<forall>q \<in> set rs. apder_nf q \<and> nonalt q \<and> q \<noteq> RZERO)"
| "apder_nf (RSEQ r1 r2) =
  (apder_nf r1 \<and> apder_nf r2 \<and> rnonseq r1 \<and> r1 \<noteq> RZERO \<and> r1 \<noteq> RONE \<
    \<and> r2 \<noteq> RZERO \<and> r2 \<noteq> RONE)"
| "apder_nf (RSTAR r) = apder_nf r"
| "apder_nf (RNTIMES r n) = apder_nf r"
| "apder_nf (RBACKREF4 r1 r2 r3 r4 cs) = True"
| "apder_nf (RHALL r cs rep) = True"
| "apder_nf (RRESIDUE cs rep) = True"

(* active/AntimirovFactoredTransition.thy:10394 *)
fun rntimes_free :: "rrexpr \<Rightarrow> bool" where
  "rntimes_free RZERO = True"
| "rntimes_free RONE = True"
| "rntimes_free (RCHAR c) = True"
| "rntimes_free (RALTS rs) = (\<forall>q \<in> set rs. rntimes_free q)"
| "rntimes_free (RSEQ r1 r2) = (rntimes_free r1 \<and> rntimes_free r2)"
| "rntimes_free (RSTAR r) = rntimes_free r"
| "rntimes_free (RNTIMES r n) = False"
| "rntimes_free (RBACKREF4 r1 r2 r3 r4 cs) = True"
| "rntimes_free (RHALL r cs rep) = True"
| "rntimes_free (RRESIDUE cs rep) = True"

(* active/AntimirovFactoredTransition.thy:31587 -- the clean-fragment guard you may ASSUME on r. *)
definition apder_clean :: "rrexpr \<Rightarrow> bool" where
  "apder_clean r \<longleftarrowrightarrow>
    legacy_rrexpr r \<and> rntimes_free r \<and> apder_nf r \<and> apder_zero_budget_trivial r"
(* legacy_rrexpr = no backref/half/residue constructors; apder_zero_budget_trivial = a minor well-
  \<rightarrow> formedness
  flag. The two that matter for the count argument are rntimes_free (RNTIMES-free) and apder_nf (normal
  \<rightarrow> form).
  So you may freely assume r is built only from RZERO/RONE/RCHAR/RSEQ/RALTS/RSTAR, in apder_nf. *)

=====
C2. AUXILIARY MEASURES & PREDICATES (referenced in the statements above/below)
=====

(* active/AntimirovFactoredTransition.thy:380 -- ||X|| : opened-set SIZE = sum of rsize over a finite set
  \<rightarrow> *)
definition rsize_set :: "rrexpr set \<Rightarrow> nat" where
  "rsize_set U = (\<Sum>q \<in> U. rsize q)"

(* base/BasicIdentities.thy:640 -- "not a top-level alternation" *)
fun nonalt :: "rrexpr \<Rightarrow> bool" where
  "nonalt (RALTS rs) = False"
| "nonalt r = True"

(* base/GeneralRegexBound.thy:3884 -- "not a top-level sequence" *)
fun rnonseq :: "rrexpr \<Rightarrow> bool" where
  "rnonseq (RSEQ r1 r2) = False"
| "rnonseq r = True"

```

```

(* active/AntimirovFactoredTransition.thy:29681 -- odfront k = open the frontier of k (NON-strong); used
  ↪ in SE *)
definition odfront :: "rrexpr \<Rightarrow> rrexpr set" where
  "odfront k = row_dlformss_set (rfrontier k)"

(* base/GeneralRegexBound.thy:50 -- classical regexes (no backref/half/residue); component of apder_clean
  ↪ *)
fun legacy_rrexpr :: "rrexpr \<Rightarrow> bool" where
  "legacy_rrexpr RZERO = True"
| "legacy_rrexpr RONE = True"
| "legacy_rrexpr (RCHAR c) = True"
| "legacy_rrexpr (RALTS rs) = (\<forall>r \<in> set rs. legacy_rrexpr r)"
| "legacy_rrexpr (RSEQ r1 r2) = (legacy_rrexpr r1 \<and> legacy_rrexpr r2)"
| "legacy_rrexpr (RSTAR r) = legacy_rrexpr r"
| "legacy_rrexpr (RNTIMES r n) = legacy_rrexpr r"
| "legacy_rrexpr (RBACKREF4 r1 r2 r3 r4 cs) = False"
| "legacy_rrexpr (RHALF r cs rep) = False"
| "legacy_rrexpr (RRESIDUE cs rep) = False"

(* active/AntimirovFactoredTransition.thy:29669 -- the "weight" = #letters + #stars. The GREEN root count
  ↪ is
  card(apder_rows r) <= apder_zw2 r + 2 <= rsize r + 2 (rntimes-free). *)
fun apder_zw2 :: "rrexpr \<Rightarrow> nat" where
  "apder_zw2 RZERO = 0"
| "apder_zw2 RONE = 0"
| "apder_zw2 (RCHAR c) = 1"
| "apder_zw2 (RALTS rs) = sum_list (map apder_zw2 rs)"
| "apder_zw2 (RSEQ r1 r2) = apder_zw2 r1 + apder_zw2 r2"
| "apder_zw2 (RSTAR r) = Suc (apder_zw2 r)"
| "apder_zw2 (RNTIMES r n) = n * Suc (apder_zw2 r)"
| "apder_zw2 (RBACKREF4 r1 r2 r3 r4 cs) = 0"
| "apder_zw2 (RHALF r cs rep) = 0"
| "apder_zw2 (RRESIDUE cs rep) = 0"

(* active/AntimirovFactoredTransition.thy:31500 -- non-triviality flag (component of apder_clean): on the
  ↪ clean
  fragment it just forces every alternation to have positive weight. NOT relevant to the injection design
  ↪ . *)
fun apder_zero_budget_trivial :: "rrexpr \<Rightarrow> bool" where
  "apder_zero_budget_trivial RZERO = True"
| "apder_zero_budget_trivial RONE = True"
| "apder_zero_budget_trivial (RCHAR c) = True"
| "apder_zero_budget_trivial (RALTS rs) =
  (apder_zw2 (RALTS rs) \<noteq> 0 \<and> (\<forall>q \<in> set rs. apder_zero_budget_trivial q))"
| "apder_zero_budget_trivial (RSEQ r1 r2) =
  (apder_zero_budget_trivial r1 \<and> apder_zero_budget_trivial r2)"
| "apder_zero_budget_trivial (RSTAR r) = apder_zero_budget_trivial r"
| "apder_zero_budget_trivial (RNTIMES r n) = False"
| "apder_zero_budget_trivial (RBACKREF4 r1 r2 r3 r4 cs) = False"
| "apder_zero_budget_trivial (RHALF r cs rep) = False"
| "apder_zero_budget_trivial (RRESIDUE cs rep) = False"

=====
D. THE CARRIER + GREEN LEMMAS TO CITE (statements; all proved, no sorry)
=====

(* cubic/DirectUniverseCubic.thy:341 -- THE CARRIER. Note strong_apder_acc r k = the strong closure of
  ↪ the
  sigma4 carrier; because rsimpStrong_dlform_closure distributes over its set, the per-constructor
  ↪ carrier
  splits below are clean SUBSETS (RSEQ/RSTAR/RCHAR). The RALTS split is the open one. *)
definition strong_apder_acc :: "rrexpr \<Rightarrow> rrexpr \<Rightarrow> rrexpr set" where
  "strong_apder_acc r k =
  rsimpStrong_dlform_closure (rfrontier (rsimp4_SEQ_atom r k) \<union> apder_term_frontier_acc r k)"

(* cubic/DirectUniverseCubic.thy:350 -- CLEAN RSEQ carrier split (GREEN) *)
lemma strong_apder_acc_RSEQ_subset:
  "strong_apder_acc (RSEQ r1 r2) k \<subseteq>

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    strong_apder_acc r1 (rsimp4_SEQ_atom r2 k) \<union> strong_apder_acc r2 k"

(* cubic/DirectUniverseCubic.thy:370 -- CLEAN RSTAR carrier split (GREEN) *)
lemma strong_apder_acc_RSTAR_subset:
  "strong_apder_acc (RSTAR r) k \<subseq>
    row_dlforms (rsimpStrong_raw (rsimp4_SEQ_atom (RSTAR r) k)) \<union>
    strong_apder_acc r (rsimp4_SEQ_atom (RSTAR r) k)"

(* cubic/DirectUniverseCubic.thy:377 -- CLEAN RCHAR carrier split (GREEN) *)
lemma strong_apder_acc_RCHAR_subset:
  "strong_apder_acc (RCHAR c) k \<subseq>
    row_dlforms (rsimpStrong_raw (rsimp4_SEQ_atom (RCHAR c) k)) \<union>
    row_dlformss_set (rsimpStrong_raw ' rfrontier k)"

(* cubic/DirectUniverseCubic.thy:385 *)
lemma strong_apder_acc_RONE_RONE [simp]: "strong_apder_acc RONE RONE = {RONE}"

(* cubic/DirectUniverseCubic.thy:504 -- the continuation-base monotonicity used in the RSEQ telescope *)
lemma strong_apder_acc_RONE_sigma_subset:
  "strong_apder_acc RONE (rsimp4_SEQ_atom r k) \<subseq> strong_apder_acc r k"

(* cubic/DirectUniverseCubic.thy:389 -- BRIDGE: U(r) sits inside the carrier at k = RONE (GREEN) *)
lemma apder_strong_dlfrontier_subset_strong_apder_acc_RONE:
  assumes "apder_nf r"
  shows "apder_strong_dlfrontier r \<subseq> strong_apder_acc r RONE"

(* cubic/DirectUniverseCubic.thy:598 -- RCHAR diff-step base (GREEN): one new row over the base *)
lemma card_strong_apder_acc_RCHAR_diff_base_le:
  assumes "apder_nf k"
  shows "card (strong_apder_acc (RCHAR c) k - strong_apder_acc RONE k) \<le> rsize (RCHAR c)"

(* cubic/DirectUniverseCubic.thy:435 -- continuation rows fall into the RONE base (GREEN) *)
lemma row_dlforms_rsimpStrong_raw_subset_strong_apder_acc_RONE:
  assumes "apder_nf k"
  shows "row_dlforms (rsimpStrong_raw k) \<subseq> strong_apder_acc RONE k"

(* cubic/DirectUniverseCubic.thy:633 -- the telescoping combinator (GREEN), core of RSEQ/RSTAR diff-steps
    ↪ *)
lemma card_Un_Diff_telescope_le:
  assumes "finite A" and "finite B" and "M \<subseq> B"
  shows "card ((A \<union> B) - C) \<le> card (A - M) + card (B - C)"

(* cubic/DirectUniverseCubic.thy:307 -- EXCESS -> absolute lift (GREEN) *)
lemma card_le_Suc_card_Diff_singleton:
  assumes "finite A"
  shows "card A \<le> Suc (card (A - {x}))"

(* cubic/DirectUniverseCubic.thy:328 (GREEN) *)
lemma card_le_rsize_set:
  assumes "finite A"
  shows "card A \<le> rsize_set A"

(* active/AntimirovFactoredTransition.thy:19053 -- the CLEAN RALTS subset for the OTHER carrier U(r) (
    ↪ GREEN).
    (Useful contrast: U is RALTS-clean but RSEQ/RSTAR-leaky; strong_apder_acc is RSEQ/RSTAR/RCHAR-clean but
    RALTS-leaky. Neither is clean on all constructors -- the cross-prune wall.) *)
lemma apder_strong_dlfrontier_RALTS_subset:
  "apder_strong_dlfrontier (RALTS rs) \<subseq> (\<Union>q \<in> set rs. apder_strong_dlfrontier q)"

(* active/AntimirovFactoredTransition.thy:37190 -- the linear ROOT count (GREEN) *)
lemma card_apder_rows_clean_le_rsize_plus_2:
  assumes "apder_clean r"
  shows "card (apder_rows r) \<le> rsize r + 2"

(* cubic/DirectUniverseCubic.thy:174 -- the GATE, GREEN modulo the CARD hypothesis. Discharge CARD and
    ↪ done. *)
lemma actual_gate_from_direct_universe_rowlevel:
  assumes "apder_clean r"

```

```

    and "card (apder_strong_dlfrontier r) \<le> Suc (rsize r)"
  shows "rsize_set (row_dlformss (rpder_strong_rows_raw c (afactored1 r s))) \<le> 2 * (rsize r + 3) ^ 3"
(* (per_row_size_le_quadratic @cubic:139 + universe_le_cubic_rowlevel @cubic:156 supply the rest; a looser
  LINEAR CARD bound card(U r) <= a*rsize r + b also closes the Gate after loosening
  ↪ budget_suc_quad_le_cube.) *)

=====
E. THE REFUTED SHORTCUT (so you don't re-propose it)
=====

(* active/AntimirovFactoredTransition.thy:29688 -- opened_boundary_forms opens with sigma4/rfrontier (NON
  ↪ -strong),
  NOT rsimpStrong_raw; odfront k = row_dlformss_set (rfrontier k). *)
definition opened_boundary_forms :: "rrexpr \<Rightarrow> rrexpr \<Rightarrow> rrexpr set" where
  "opened_boundary_forms r k =
    row_dlformss_set (rfrontier (rsimp4_SEQ_atom r k) \<union> apder_term_frontier_acc r k) - odfront k"

(* active/AntimirovFactoredTransition.thy:32290 -- GREEN proof that U(r) does NOT embed into the non-
  ↪ strong
  opened_boundary_forms (so the green card bound on opened_boundary_forms does NOT transfer to U): *)
lemma rsimpStrong_dlform_closure_opened_boundary_carrier_false:
  fixes a :: char
  defines "r \<equiv> RSTAR (RALTS [RCHAR a])"
  defines "bad \<equiv> RSTAR (RCHAR a)"
  shows "apder_clean r"
    and "bad \<in> rsimpStrong_dlform_closure (set (afactored1 r []))"
    and "bad \<notin> odfront RONE \<union> opened_boundary_forms r RONE"
    and "\<not> rsimpStrong_dlform_closure (set (afactored1 r [])) \<subsepeq>
      odfront RONE \<union> opened_boundary_forms r RONE"

```